

# Interpretable Road Damage Detection for Intelligent Transportation Systems using YOLOv8 Nano and EigenCAM

## Technical Report

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## Abstract

Road damage detection is an important application of computer vision for intelligent transportation systems, enabling faster and more consistent infrastructure inspection than traditional manual surveys. However, conventional object detection models often operate as black-box systems, making it difficult to understand the reasoning behind their predictions.

This project presents an interpretable deep learning framework for automated road damage detection using the YOLOv8 Nano object detection model. A custom dataset containing three road damage categories—cracks, potholes, and manholes—was used to train and evaluate the detector. After 30 training epochs, the trained model achieved a precision of 0.497, recall of 0.475, mAP@0.50 of 0.442, and mAP@0.50:0.95 of 0.193. Among the evaluated classes, potholes achieved the strongest detection performance, while cracks and manholes remained more challenging because of greater visual variability and class imbalance.

To improve model interpretability, EigenCAM was integrated into the inference pipeline to visualize the feature representations contributing to the detector's predictions. Multiple intermediate feature layers were evaluated, with Layer -2 producing the clearest localization around road damage while suppressing unrelated background regions. The

explainability analysis demonstrates how feature-based visualization can complement lightweight object detection by providing qualitative insight into the model's learned representations.

Overall, this work establishes a practical foundation for combining efficient road damage detection with Explainable Artificial Intelligence (XAI) techniques for intelligent infrastructure monitoring.

# 1. Introduction

Road transportation networks are critical components of modern infrastructure, supporting economic activity, public safety, and daily mobility. Over time, road surfaces deteriorate because of traffic loads, environmental conditions, and aging materials, leading to defects such as cracks, potholes, and damaged manholes. If these defects are not detected and repaired in a timely manner, they can increase maintenance costs, reduce driving comfort, damage vehicles, and contribute to road accidents.

Traditional road inspection methods primarily rely on manual surveys performed by trained personnel. Although manual inspection can provide accurate assessments, it is labor-intensive, time-consuming, expensive, and subject to human variability. As transportation networks continue to expand, there is an increasing need for automated inspection systems capable of detecting road damage efficiently and consistently.

Recent advances in deep learning, particularly convolutional neural networks (CNNs) and object detection models such as the YOLO (You Only Look Once) family, have significantly improved the ability of computer vision systems to detect objects in real time. Among these models, YOLOv8 offers an effective balance between detection accuracy and computational efficiency, making it suitable for applications requiring near real-time performance.

Despite these advances, most object detection systems provide only prediction outputs without explaining how those predictions are generated. This lack of transparency can reduce user confidence, particularly in safety-critical applications involving public infrastructure. Explainable Artificial Intelligence (XAI) techniques such as Gradient-weighted Class Activation Mapping (Grad-CAM) address this limitation by highlighting the image regions that most strongly influence a model's predictions.

Accordingly, this project develops an interpretable road damage detection framework using YOLOv8 Nano for automated object detection and EigenCAM for feature-based explainability. The proposed pipeline combines efficient road damage detection with

qualitative visualization of learned feature representations, contributing toward more transparent AI systems for intelligent infrastructure monitoring.

## 2. Related Work

Automated road damage detection has become an active research area within intelligent transportation systems due to its potential to reduce the cost and time associated with manual road inspections. Recent advances in deep learning-based object detection, particularly the YOLO family of models, have significantly improved the accuracy and efficiency of pavement defect detection (Guo & Zhang, 2022; Sami et al., 2023).

Guo and Zhang (2022) proposed an improved YOLOv5-based approach for road damage detection, demonstrating enhanced localization performance for multiple pavement defects. Subsequent studies further refined YOLO architectures by introducing lightweight network modifications, attention mechanisms, and feature fusion techniques to improve detection accuracy while maintaining real-time inference capability (Miao et al., 2023; Lightweight Model for Pavement Defect Detection Based on Improved YOLOv7, 2023; RDD-YOLO, 2024).

Recent research has also focused on comparing successive generations of YOLO models and evaluating their suitability for practical deployment in intelligent transportation systems. These studies consistently report that lightweight object detectors provide an effective balance between computational efficiency and detection performance, making them attractive candidates for edge deployment and real-time infrastructure monitoring (Optimizing YOLO Architectures, 2024; Towards Real-world Deployment of Deep Learning Solutions for Road Damage Detection, 2024).

Although modern object detection models achieve promising detection accuracy, they generally operate as black-box systems, providing limited insight into the reasoning behind their predictions. This limitation has motivated growing interest in Explainable Artificial Intelligence (XAI). Gradient-weighted Class Activation Mapping (Grad-CAM) has emerged as one of the most widely adopted explainability techniques for visualizing the image regions that influence deep learning predictions. Recent studies have demonstrated that Grad-CAM can be successfully applied to object detection models, including YOLO-based detectors, to improve model transparency and facilitate error analysis (Kirchknopf et al., 2022; Cheng et al., 2025).

Building upon these previous studies, this project combines lightweight object detection using YOLOv8 Nano with EigenCAM-based explainability. Unlike conventional object detection pipelines that only provide bounding-box predictions, the proposed approach incorporates feature visualization to improve model interpretability while maintaining

computational efficiency suitable for intelligent transportation and infrastructure monitoring applications.

### 3. Dataset

The experiments in this project were conducted using a road damage dataset containing three object categories:

- Crack
- Manhole
- Pothole

The dataset consists of a total of 2,009 annotated images divided into training, validation, and testing subsets.

**Table 1. Dataset Statistics**

| Dataset Split | Number of Images |
|---------------|------------------|
| Training      | 1406             |
| Validation    | 301              |
| Testing       | 302              |
| <b>Total</b>  | <b>2009</b>      |

The dataset contains images captured under diverse road conditions, illumination levels, and viewing perspectives. Such variability increases the difficulty of the detection task while improving the model’s ability to generalize to unseen road scenes.

### 4. Methodology

The overall workflow of this project consists of dataset preparation, model training, performance evaluation, and explainability analysis. The objective is to develop a lightweight object detection system capable of accurately identifying common types of road damage while providing interpretable predictions.

#### 4.1 Data Preparation

The dataset was divided into training, validation, and testing subsets. Each image contains bounding box annotations corresponding to one of three road damage

categories: crack, manhole, and pothole. The dataset configuration was provided to the YOLOv8 training framework through a YAML configuration file.

## 4.2 Model Selection

YOLOv8 Nano (YOLOv8n) was selected as the baseline object detection model because of its lightweight architecture, fast inference speed, and suitability for real-time applications. Although larger YOLO models generally provide higher detection accuracy, the Nano variant offers an effective balance between computational efficiency and performance.

## 4.3 Model Training

The model was trained for 30 epochs using the Ultralytics YOLOv8 framework on a Tesla T4 GPU in Google Colab. During training, the model optimized its weights using annotated road damage images while monitoring validation performance after each epoch.

## 4.4 Performance Evaluation

Model performance was evaluated using commonly adopted object detection metrics including Precision, Recall, mean Average Precision at IoU threshold 0.50 (mAP@0.50), and mean Average Precision across IoU thresholds from 0.50 to 0.95 (mAP@0.50:0.95). Confusion matrices and precision-recall curves were also analyzed to understand class-wise performance.

## 4.5 Explainability Analysis

To improve the interpretability of the trained YOLOv8 Nano detector, EigenCAM was applied to visualize the learned feature representations responsible for road damage detection. Unlike gradient-based explainability methods, EigenCAM generates activation maps directly from feature maps using principal component analysis (PCA), making it computationally efficient while avoiding gradient computation.

A representative pothole image from the test dataset was selected for qualitative analysis. The trained YOLOv8 Nano detector first generated the object detection prediction, after which EigenCAM was applied to visualize the regions that contributed most strongly to the model's learned feature representations.

To identify the most informative feature representations, multiple intermediate feature layers (-2, -3, -4, and -5) were evaluated. The activation maps produced by these layers

were visually compared based on localization quality and background suppression.

Among the evaluated layers, Layer -2 consistently produced the clearest localization around the road surface while minimizing activations in unrelated background regions. Deeper layers generated increasingly sparse feature maps and occasionally emphasized surrounding structures rather than the damaged pavement itself. Consequently, Layer -2 was selected for the final explainability analysis presented in this report.

The qualitative results demonstrate that EigenCAM effectively highlights the image regions that contribute most strongly to the detector's internal feature representations, providing additional transparency beyond conventional bounding-box predictions.

### **Original Test Image**

Figure 1. Original test image selected for qualitative explainability analysis.

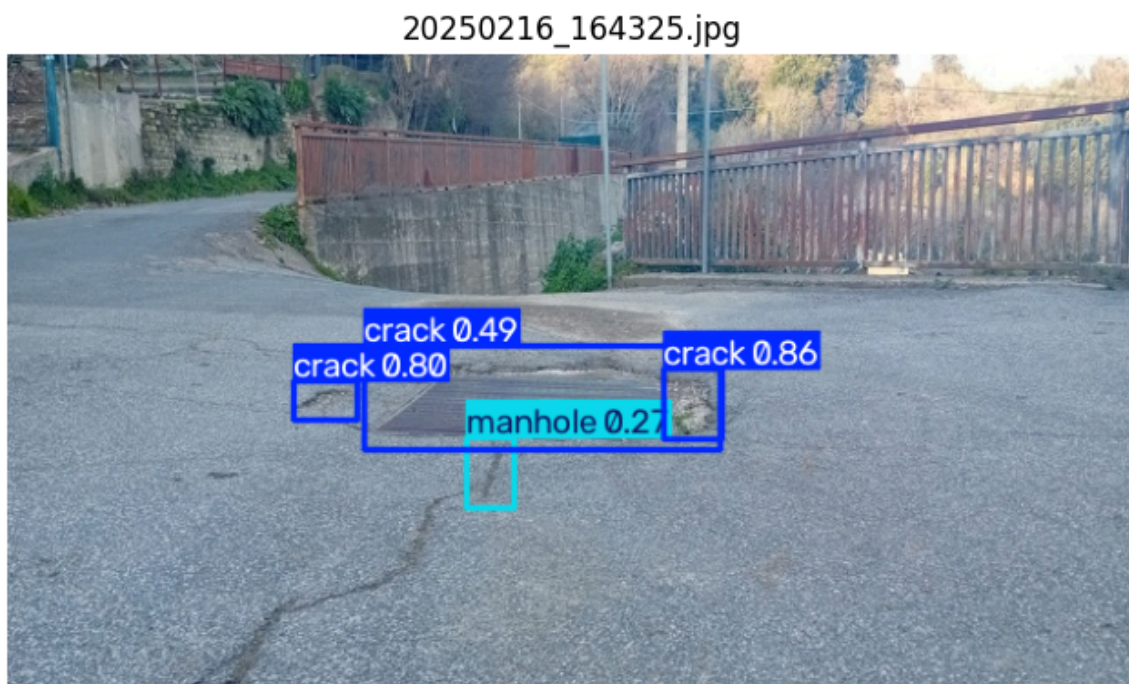
#### **Selected Test Image**



### **YOLOv8 Nano Detection**



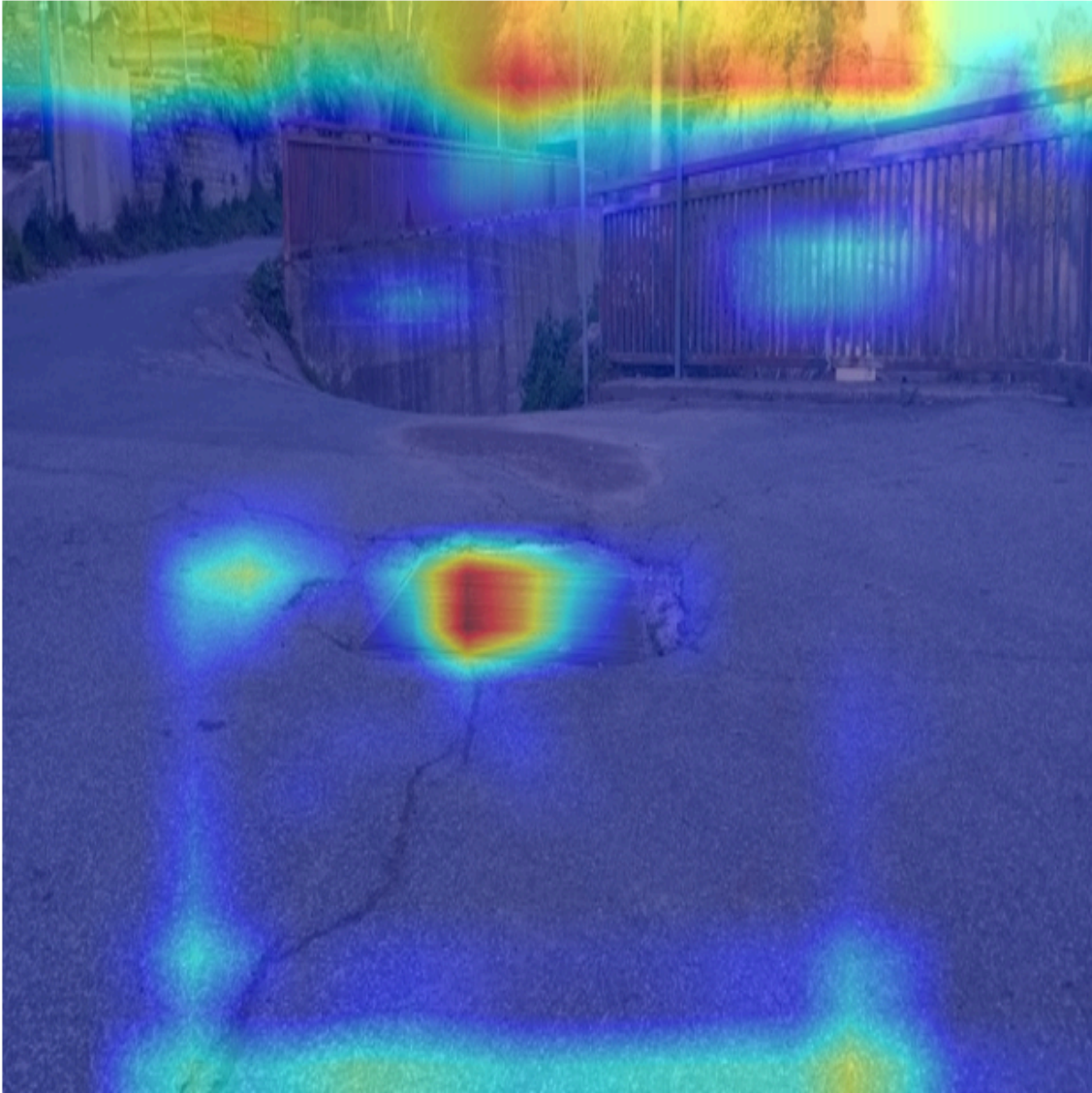
Figure 2. YOLOv8 Nano detection showing pothole localization on the selected test image.



### EigenCAM Visualization

Figure 3. EigenCAM visualization highlighting the feature representations contributing to the detector's prediction.

EigenCAM (-2)



Figures 1–3 illustrate the complete explainability workflow adopted in this project. The original road image is first processed by the trained YOLOv8 Nano detector, which successfully localizes the pothole using a bounding box. EigenCAM is subsequently applied to visualize the feature representations responsible for the detector’s prediction. The resulting activation map demonstrates that the detector primarily focuses on the damaged pavement region while suppressing unrelated background areas, providing qualitative evidence that the learned features correspond closely to the detected road defect.

## 5. Experimental Setup

The experiments were conducted using the Ultralytics implementation of YOLOv8 within the Google Colab environment. Model training was performed using an NVIDIA Tesla T4 GPU, enabling efficient optimization of the lightweight YOLOv8 Nano architecture.



The detector was trained for 30 epochs using images resized to  $640 \times 640$  pixels. The dataset was divided into predefined training, validation, and testing subsets consisting of 1,406, 301, and 302 images, respectively. The default optimization strategy provided by the Ultralytics framework was used throughout training.

Model performance was monitored using validation metrics generated after each training epoch. Precision, Recall, mAP@0.50, and mAP@0.50:0.95 were selected as the primary evaluation metrics because they are widely adopted for benchmarking object detection models. In addition, confusion matrices, precision–recall curves, and F1-score curves were analyzed to better understand class-wise performance and model behavior.

## 6. Results

The trained YOLOv8 Nano model successfully learned to detect three categories of road damage: cracks, manholes, and potholes. Performance was evaluated using the validation dataset after 30 training epochs.

### 6.1 Overall Detection Performance

The final validation metrics obtained from the trained model are summarized in **Table 2**.

**Table 2. Overall Validation Performance**

| Metric        | Value |
|---------------|-------|
| Precision     | 0.497 |
| Recall        | 0.475 |
| mAP@0.50      | 0.442 |
| mAP@0.50:0.95 | 0.193 |

These results demonstrate that the lightweight YOLOv8 Nano model was able to learn meaningful representations of road damage despite its limited model capacity and a relatively short training schedule of 30 epochs.

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**Figure 4. Training results across 30 training epochs.**

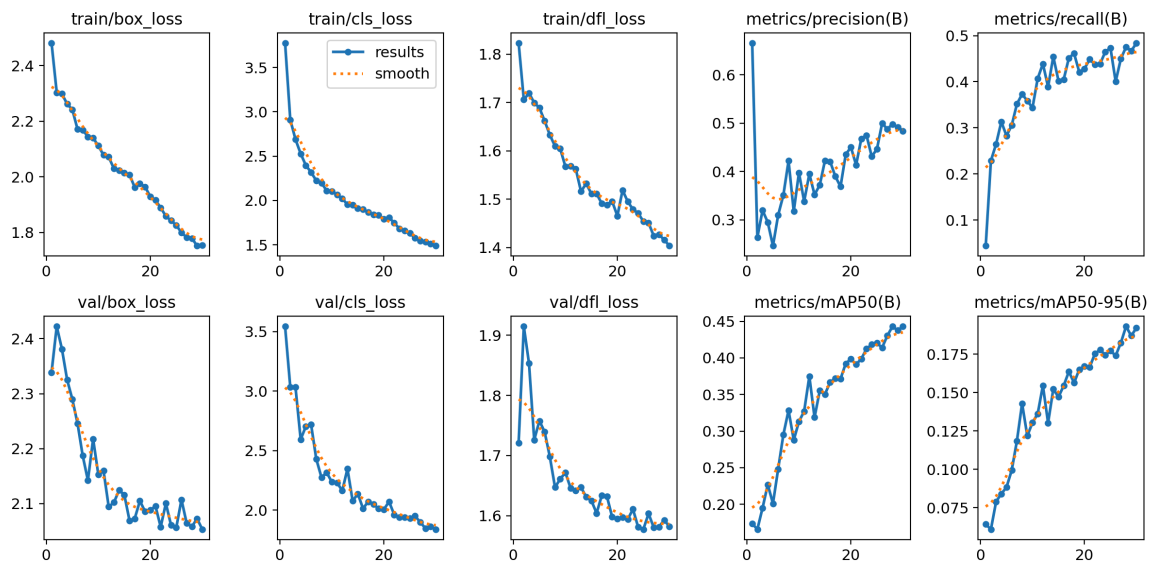


Figure 4 shows the training and validation metrics across all 30 epochs. The curves indicate stable convergence with no evidence of severe overfitting.

*Training metrics over 30 epochs showing the convergence behavior of the YOLOv8 Nano model.*

## 6.2 Class-wise Performance

Performance varied across the three road damage categories.

| Class   | mAP@0.50 |
|---------|----------|
| Crack   | 0.360    |
| Manhole | 0.309    |
| Pothole | 0.658    |

Among all classes, potholes achieved the highest detection accuracy. Their larger size and more distinctive visual appearance likely contributed to improved recognition performance. In contrast, cracks and manholes proved more challenging because they often exhibit irregular shapes, varying illumination conditions, and visual similarities to surrounding road surfaces.

**Figure 5. Precision–Recall curve of the trained YOLOv8 Nano detector.**

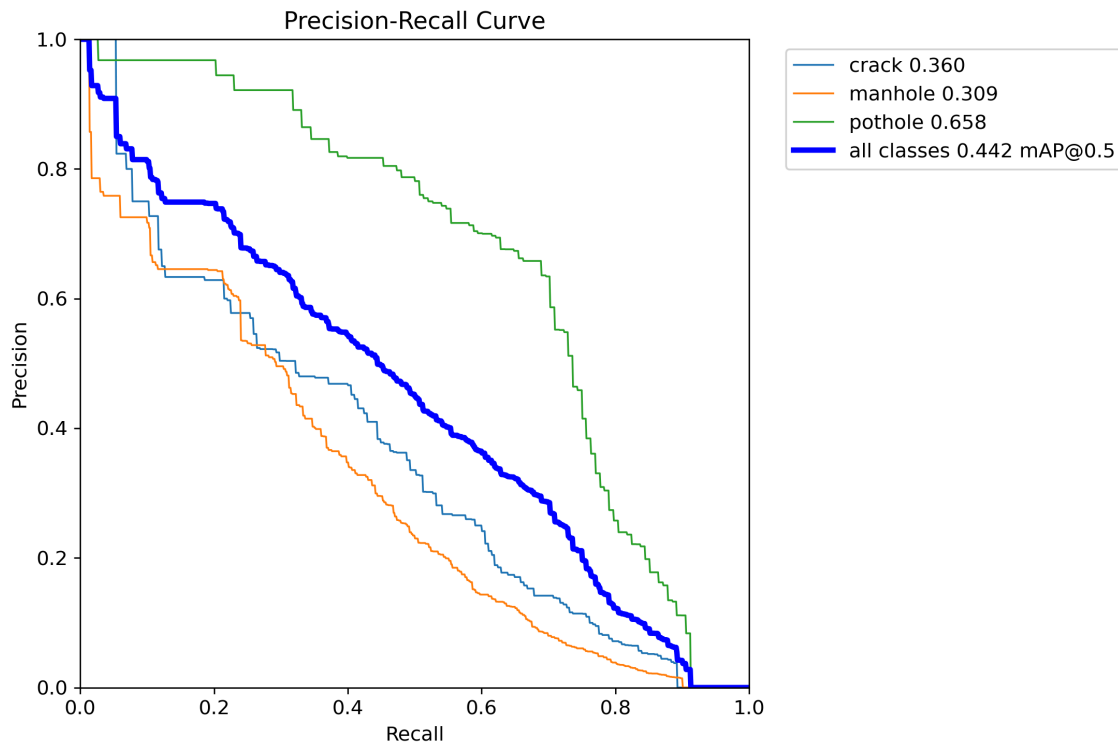


Figure 5 presents the Precision–Recall curves for the three road damage categories. Potholes achieved the highest detection performance, while cracks and manholes remained comparatively more challenging.

*Precision–Recall curve illustrating the trade-off between precision and recall across confidence thresholds.*

## 6.3 Training Behaviour

The training process showed gradual improvements in localization and classification performance over successive epochs. Precision–Recall curves, F1-score curves, and confusion matrices indicate that the detector learned meaningful representations of road damage while still exhibiting confusion between visually similar classes.

The training visualizations provide additional insight into the detector’s learning behaviour, illustrating the evolution of precision, recall, and class-wise confusion throughout the training process.

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**Figure 6. Confusion matrix illustrating per-class detection performance.**

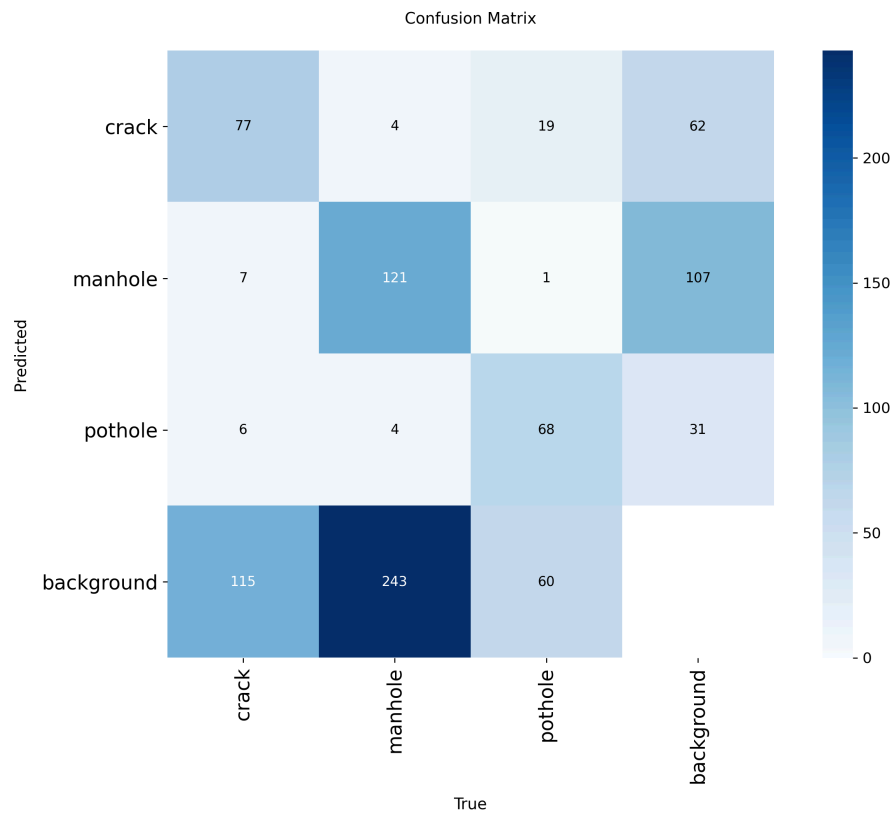


Figure 6 illustrates the confusion matrix of the trained detector. Most prediction errors occurred between crack and manhole categories, whereas potholes were classified more reliably.

*Confusion matrix showing class-wise prediction performance on the validation dataset.*

## 6.4 Prediction Examples

Qualitative evaluation of the trained model demonstrated successful detection of multiple road damage categories across different road scenes. Representative prediction examples have been included in the repository, covering successful detections of cracks, potholes, and manholes, as well as challenging failure cases. Including both successful and unsuccessful predictions provides a more comprehensive assessment of the detector’s strengths and current limitations.

## 7. Discussion

The experimental results demonstrate that YOLOv8 Nano can effectively detect road damage while maintaining a lightweight architecture suitable for real-time applications. Despite being trained for only 30 epochs, the detector successfully learned meaningful visual representations for cracks, potholes, and manholes.

Among the evaluated classes, potholes achieved the highest detection performance ( $\text{mAP}@0.50 = 0.658$ ), whereas cracks and manholes remained more challenging because of their irregular appearance, varying illumination conditions, and similarity to surrounding pavement textures. Qualitative prediction examples further revealed that water-filled potholes, overlapping defects, and complex road backgrounds occasionally resulted in incorrect classifications despite successful localization.

Beyond quantitative evaluation, EigenCAM provided valuable qualitative insight into the detector's learned feature representations. The activation maps demonstrated that the model primarily focused on damaged pavement regions while suppressing unrelated background structures. The comparison of multiple intermediate feature layers further showed that Layer -2 produced the clearest localization, suggesting that intermediate representations provide a favorable balance between semantic information and spatial detail.

Overall, the results indicate that combining lightweight object detection with feature-based explainability improves the transparency of deep learning models and provides a stronger foundation for future intelligent infrastructure monitoring systems.

## 8. Limitations

Although the proposed framework demonstrates promising performance, several limitations remain.

The detector was trained for only 30 epochs, and additional training may further improve convergence and detection accuracy. Furthermore, only the YOLOv8 Nano architecture was evaluated; larger YOLO variants may achieve higher performance at the expense of increased computational cost.

The dataset also contains class imbalance, varying illumination conditions, complex road textures, and subtle crack patterns that continue to challenge the detector. These factors contribute to missed detections and incorrect classifications, particularly for crack and manhole categories.

Finally, the explainability analysis presented in this work is qualitative. Although EigenCAM provides meaningful visualization of learned feature representations, future studies should incorporate quantitative explainability evaluation metrics to systematically assess the quality of visual explanations.

## 9. Future Work

Several directions can further improve this work.



Future research will compare EigenCAM with other explainability techniques such as Grad-CAM, Grad-CAM++, Score-CAM, and LayerCAM to better understand their suitability for object detection models.

Additional improvements include training larger YOLOv8 variants, increasing the number of training epochs, performing hyperparameter optimization, applying stronger data augmentation strategies, and evaluating additional road damage datasets.

In the longer term, the framework can be extended for real-time video-based road inspection and deployment on edge devices for intelligent transportation and smart infrastructure monitoring.

## 10. Conclusion

This project presented an interpretable framework for automated road damage detection using YOLOv8 Nano and EigenCAM. The trained detector successfully identified three categories of road damage—cracks, potholes, and manholes—and demonstrated that lightweight object detection models can provide an effective foundation for intelligent road inspection.

Beyond detection performance, the integration of EigenCAM enabled qualitative visualization of the learned feature representations responsible for model predictions. The layer comparison experiments further showed that intermediate feature representations (Layer -2) provided the most informative activation maps, improving the transparency of the detection pipeline.

Overall, this work demonstrates that combining efficient object detection with explainable artificial intelligence techniques can improve both the interpretability and practical applicability of computer vision systems for intelligent transportation and infrastructure monitoring. The proposed framework provides a solid foundation for future research in explainable road damage detection and Edge AI-enabled infrastructure inspection.

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